DCIT 26

Emerging Technologies and Their Areas

A Short Report on Emerging Technologies and Their Uses

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Date:	September 12, 2026

1. What Are Emerging Technologies?

Emerging technologies are new or developing tools and systems that may bring important changes to people, organizations, and society. A technology can also be considered emerging when an older invention continues to improve or is used in a new way. Artificial intelligence (AI) is one example. Although research on AI began many years ago, it is still developing and is now used in areas such as education, healthcare, business, and research.

2. Five Characteristics of Emerging Technologies

Rotolo, Hicks, and Martin (2015) identify five features that can help describe an emerging technology:

Radical novelty: It offers a new idea or a different way to use existing technology. AI, for example, can be applied to many kinds of tasks.

Relatively fast growth: It may develop, improve, or spread quickly compared with more established technologies.

Coherence: Different related tools or developments work together and form a connected area of technology.

Prominent impact: It can influence several parts of society, including education, business, health, and everyday life.

Uncertainty and ambiguity: Its future uses and possible effects may not be completely clear yet.

3. Importance to IT Students

Learning about emerging technologies helps IT and computer science students prepare for changes in the technology industry. Subjects such as AI, machine learning, the Internet of Things (IoT), cloud computing, big data, quantum computing, augmented reality, and virtual reality introduce students to tools that may be used in future jobs. Knowing about these areas also helps students understand new developments and continue building their skills.

4. Effects on Society

New technologies can change how people study, communicate, work, and complete daily activities. When used properly, they can save time and make some tasks easier. AI, for example, can assist workers and organizations in education, healthcare, and business. However, people should also consider the possible problems and risks that may come with these tools.

5. Augmented Reality and Virtual Reality

Augmented Reality (AR) adds digital content, such as images, words, sounds, or effects, to a person's view of the real world.

Virtual Reality (VR) uses a computer to create a three-dimensional environment that people can explore and interact with.

6. Advantages and Disadvantages of Artificial Intelligence

Advantages

• **Less repetitive work:** AI can perform routine tasks that would otherwise take people time and effort.

• **Can operate for long periods:** Some AI systems can run continuously when they are properly maintained.

• **Supports new developments:** AI can help create new tools and ways of solving problems.

• **Useful in many fields:** It can be applied in education, healthcare, business, and other areas.

• **May improve productivity:** It can speed up certain tasks and help organize work.

Disadvantages

• **Possible job changes:** Automation may reduce the need for some jobs or require workers to learn new skills.

• **Too much dependence:** Overusing AI may reduce opportunities to practice independent thinking.

• **Creative limitations:** AI works from learned patterns and may not understand ideas in the same way people do.

• **Privacy concerns:** Handling personal or sensitive information can create security risks.

• **Incorrect results:** AI can make mistakes, so its output should be checked.

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DCIT 26 - Applications Development and Emerging Technologies. Module 1, Week 2: Emerging Technologies Overview and Its Areas.